



HerWILL × UAP Datathon 2026

DETECTING EXPLICIT AND SUBTLE **HARMFUL CONTENT** IN BANGLA, ENGLISH, AND BANGLISH SOCIAL MEDIA

Team : IUT_BakeNekos 【化け猫】

Team Members : Md. Farhan Ishraq, Didhiti Nahid, Tamim Rayeed

Date : 1st October, 2026

Introduction

Problem Statement :

- Classify Bangla, English, and Banglish social media text into Explicitly Harmful, Subtly Harmful, or Non-Toxic/Neutral categories.

Brief Description of our interpretation :

- **Task:** Multi-class text classification for harmful content detection.
- **Key challenge:** Detecting not only obvious harmful language, but also subtle toxicity, sarcasm, coded language, and mixed text.
- **Goal:** Build a model that generalizes well to noisy, real-world social media content.
- **Dataset:** Approximately 60K manually annotated social media posts.

Evaluation Criteria :

$$F_1^{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \frac{2P_c R_c}{P_c + R_c}$$

C = number of classes,

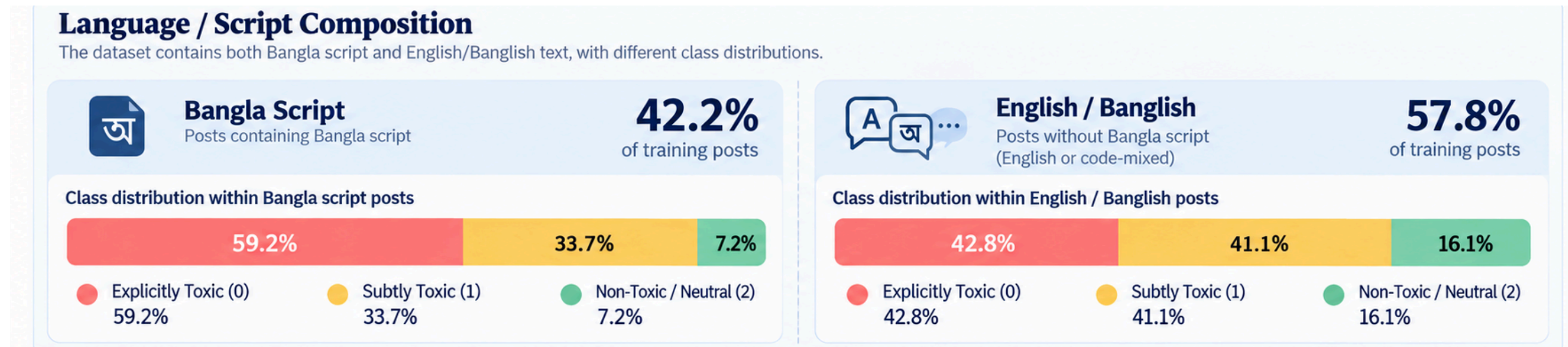
P_c = precision for class c ,

R_c = recall for class c .

Label	Category	Meaning
0	Explicitly Toxic / Harmful	Direct, overt, or clearly abusive, hateful, or harmful content.
1	Subtly Toxic / Harmful	Indirect, subtle, sarcastic, coded, or context-dependent harmful content.
2	Non-Toxic / Neutral	Content that does not contain harmful or toxic intent.

Constraints: NO EXTERNAL DATASET / 25 hours only

Dataset Characteristics Defining Our Approach



Significant class imbalance

- The minority class represents only 12.3% of the training data, motivating Macro F1 rather than accuracy as the primary metric.

Strong multilingual nature

- The dataset contains substantial Bangla and English content, with a smaller mixed-script component, motivating multilingual embeddings and multilingual LLMs.

Short and noisy social-media-style text

- With a median of only 8 words, lexical patterns, character-level information, punctuation and semantic representations are all potentially informative.

Complementary representations are valuable

- The dataset combines lexical, multilingual, stylistic and semantic variation, motivating an ensemble/stacking strategy rather than relying on a single representation.

Related Works Defining Our Approach

Harmful and implicit toxicity.

- Early work separated hate speech from merely offensive language [1].
- Later benchmarks target implicit and coded hate [2] showing classifiers can encode annotator and dialect bias [3, 4]. Our class 1 (subtle toxicity) is closest to implicit hate.

Bangla and code-mixed NLP.

- Monolingual and regional encoders such as BanglaBERT [7], MuRIL [8] and IndicBERTv2 [9] were built for Bangla and other Indic languages.
- Bangla hate-speech detection has been studied with such encoders [5], and code-switched text poses its own difficulties [6].

Multilingual encoders and LLM fine-tuning.

- Our encoder baselines are XLM-R [10], its Twitter-adapted variant XLM-T [11], mDeBERTa-v3 [12] and multilingual E5 [13]. For decoder LLMs we use low-rank adaptation [14] on 4-bit quantised weights [15], applied to Qwen3 [16] and Gemma 4 [17], and run prompted baselines with vLLM [18]. All models are trained with Hugging Face Transformers [19].

Ensembles and semi-supervision.

- Stacked generalisation [20] trains a second-level model on out-of-fold predictions of first-level models.
- We use gradient-boosted trees [21] as the stacker and compare implementations [22, 23, 24, 25].

Pre Processing & Model Selection

Text preprocessing:

- Cleaned and normalized the text while preserving meaningful linguistic information, including Bangla, English, Banglish, emojis, and punctuation.

Feature preparation:

- Generated model-specific representations such as TF-IDF features and multilingual embeddings, while *avoiding ID-based features* to prevent reliance on dataset identifiers.

We decided to use multiple complementary text models

Gemma 4 31B + QLoRA → strong semantic/LLM understanding

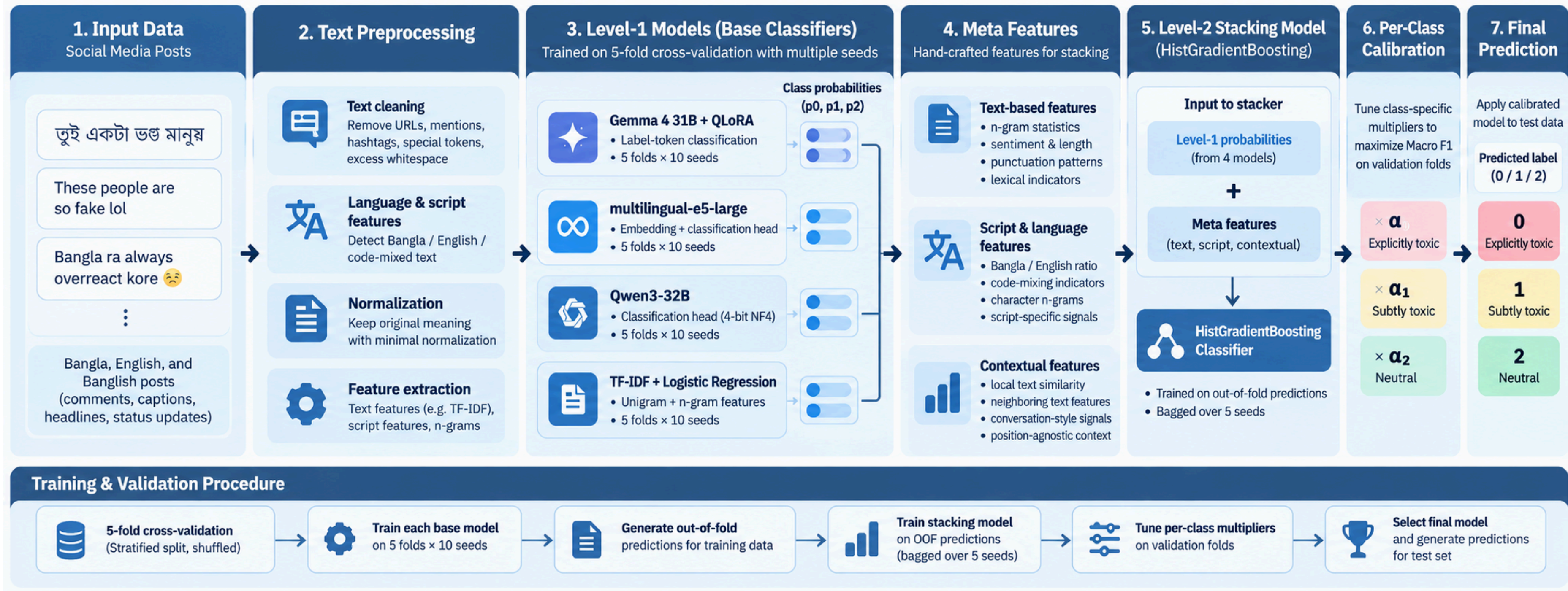
Qwen3-32B + QLoRA → another large-model perspective

multilingual-e5-large → multilingual semantic representation

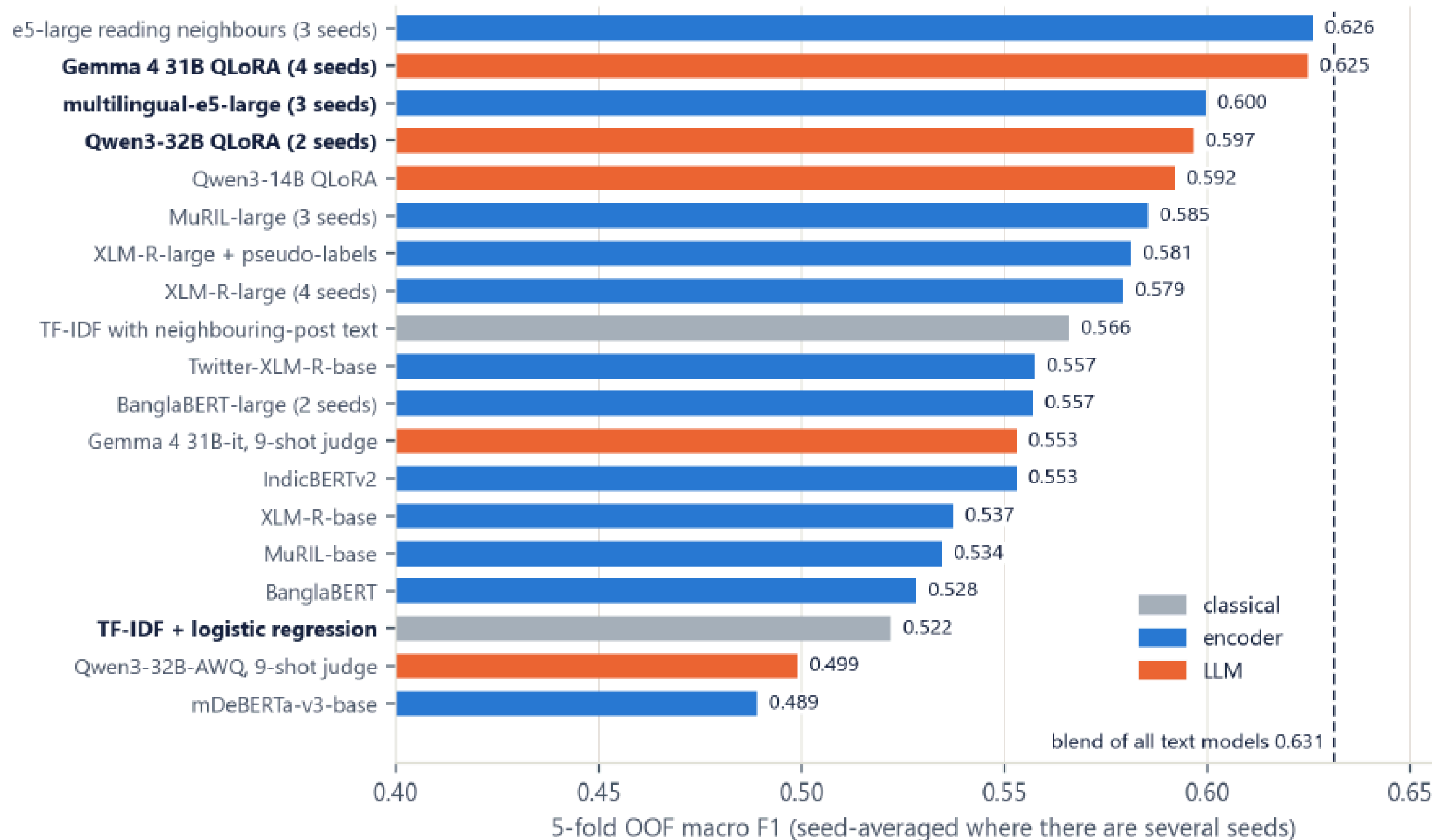
TF-IDF + Logistic Regression → strong lexical baseline

Methodology

A Gradient-boosting stacker over four fine-tuned text models on the competition dataset.

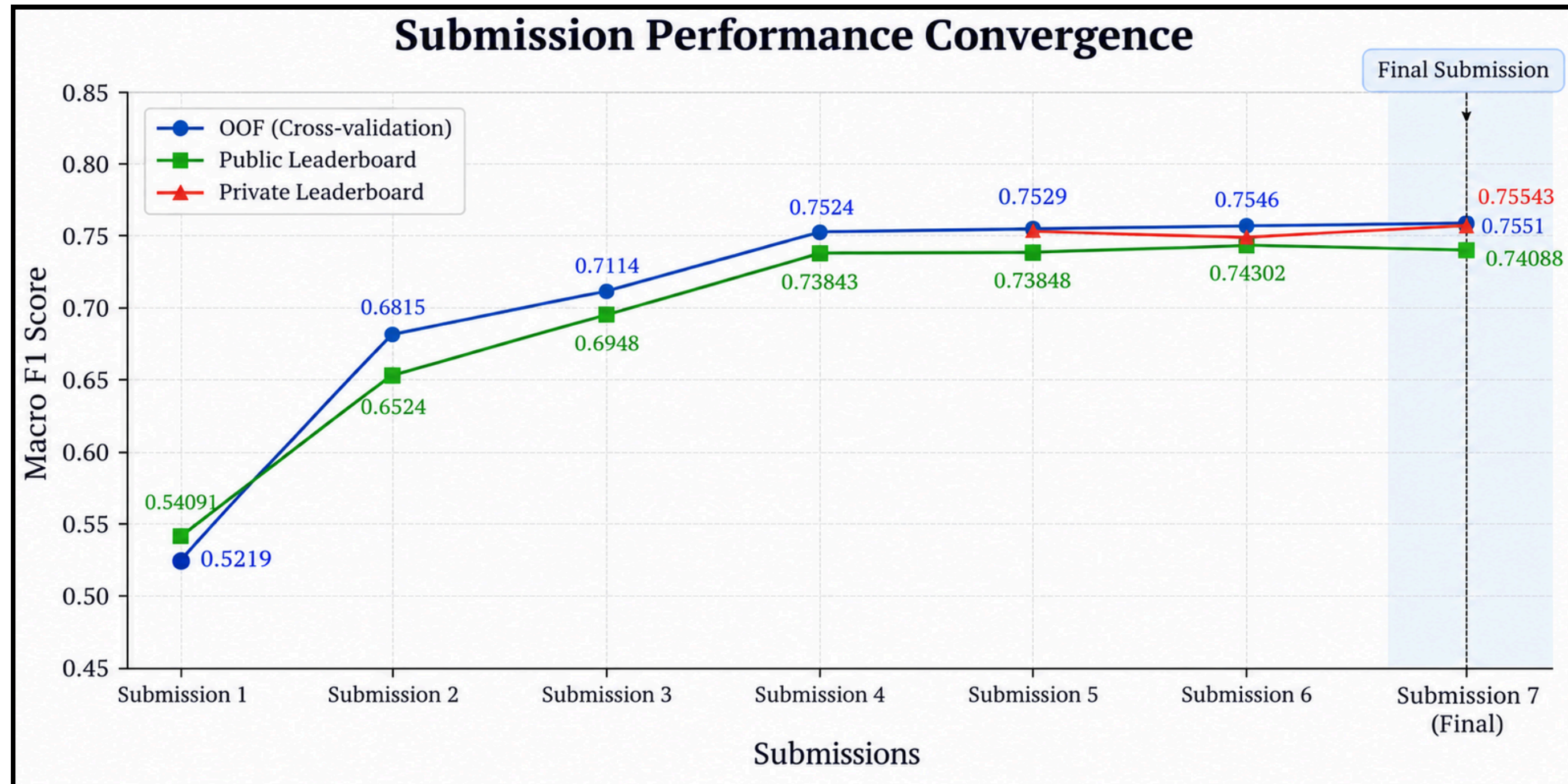


Key Results & Findings



Level-1 models ranked by OOF macro F1. The dashed line marks the text-only blend of all models (0.631).

Key Results & Findings



We prioritized cross-validated Macro F1 over short-term public leaderboard gains, focusing on robust generalization rather than leaderboard optimization.

Key Results & Findings

- The collection-order signal helps Bangla disproportionately. So the structural features reduce the language gap considerably.
- Model diversity mattered more than model count: adding more models did not guarantee better generalization.
- Cross-validation was more trustworthy than the public leaderboard.
- We selected models by cross-validation, not by short-term public-LB gains; CV predicted the final private score within 0.0003 difference. Thus, Generalization was ensured.

With text probabilities alone:

Bangla: 0.522
English/Banglish: 0.675

With the full stack:

Bangla: 0.719
English/Banglish: 0.761

Public Private

This leaderboard is calculated with approximately 40% of the test data. The final results will be based on the other 60%, so the final standings may change.

#	Team	Members	Score
1	FatWiz		0.77874
2	CUET_AL_Masaar_56		0.74414
3	IUT_BakeNekos 【化け猫】		0.74302

Your latest submission scored 0.74088. Your best score remains 0.74302.

Public Private

The private leaderboard is calculated with approximately 60% of the test data. This competition has completed. This leaderboard reflects the final standings.

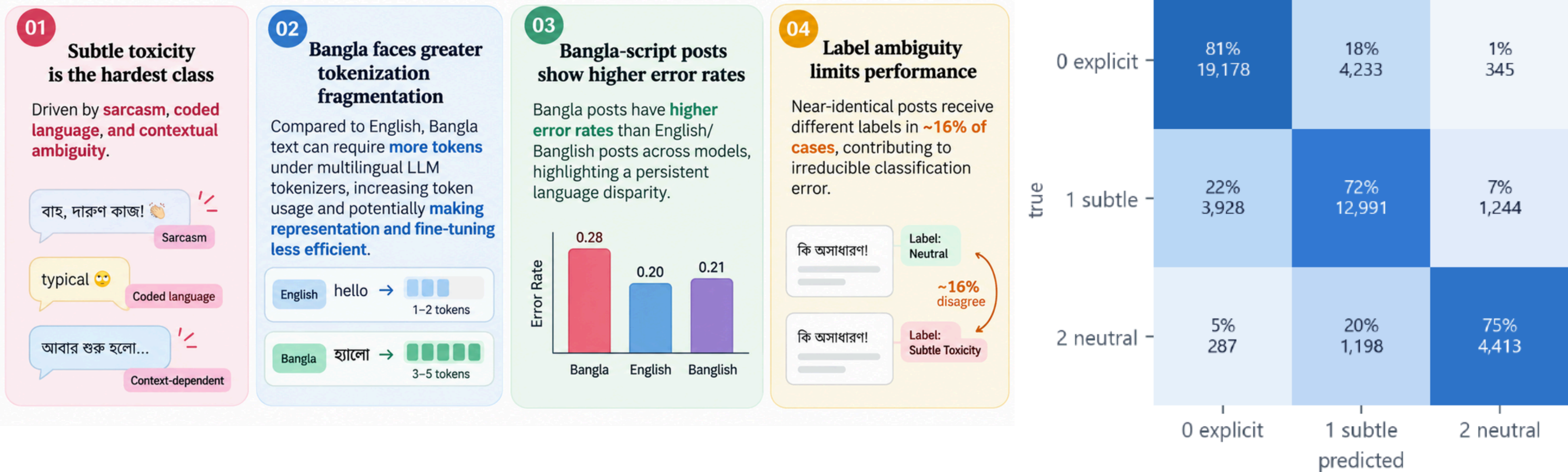
#	△	Team	Members	Score
1	—	FatWiz		0.77514
2	▲ 1	IUT_BakeNekos 【化け猫】		0.75543
3	▼ 1	CUET_AL_Masaar_56		0.75113

Error Analysis

Table 12: Per-class OOF results of the final system (after calibration). Accuracy is 0.765.

Class	Precision	Recall	F1	Support
0 explicit	0.820	0.807	0.814	23,756
1 subtle	0.705	0.715	0.710	18,163
2 neutral	0.735	0.748	0.742	5,898
Macro average	0.753	0.757	0.755	47,817

Confusion matrix of the final system on OOF predictions



THANK YOU

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